

14X065

Digital forensics

CC1 Questions

1. Principles of digital forensics
2. Digital multimedia forensics: definition, classification.
3. Active and passive approaches for multimedia authentication.
4. Image generation pipeline
5. Camera identification based on photo response non-uniformity (PRNU).
6. Media forensics based on in-camera fingerprints.
7. Application of geometrical fingerprints in digital forensics.
8. Out-camera fingerprints in digital forensics.
9. Principles of image JPEG compression.
10. JPEG compression: general compression scheme
11. JPEG compression: sources of redundancies
12. JPEG compression: particularities of luminance and chrominance channels processing
13. JPEG compression artefacts
14. Detection of double JPEG compression
15. Image editing forensics: inconsistency in light
16. Image editing forensics: local filtering traces
17. Photo manipulation with inconsistent shadows: basic assumptions
18. Photo manipulation with inconsistent shadows: cast shadow
19. Photo manipulation with inconsistent shadows: attached shadow
20. Photo manipulation with inconsistent shadows: mathematical model
21. Traces of image editing: resampling detection
22. Traces of image editing: detection of global contrast enhancement
23. Traces of image editing: detection of local contrast enhancement